

Examiner reports

Q1.

This question required the use of the exponential function and natural logs which produced a diverse range of responses. In part (a) about a third of students were able to correctly write down the required two equations using the information given in the question, but solving them simultaneously proved beyond many students. Some used the given value of b to find the value of a , which gained 1 mark and it also enabled them to tackle later parts of the question.

There were many correct responses to part (b) which was well done as long as students had found the correct values of a and b .

In part (c) students mostly failed to use inequalities but this approach did not lose marks. Those students who isolated the exponential term before taking logs were usually more successful than those who took logs straight away, as errors occurred later when simplifying the log terms. Many lost the final mark because they did not quote a year as their final answer as required. Appropriate numerical approaches were accepted and there was some follow through of the students value of ' a ', provided $b = 90$ was used.

There were some very good responses to part (d) and over half of all students were able to give a limitation of the model. It was disappointing that over 17% of students did not attempt this question despite it not being dependent upon any of the earlier work in the question.

Q2.

A high proportion of students selected the correct response for this question. Incorrect responses were split between the available options fairly evenly.

Q3.

Part (a) was completed successfully by the majority of students. It was pleasing to see students being given relatively little structure and demonstrating that they could piece together the given information to solve the problem.

In part (b), around a third of students found difficulty in translating the problem given in context into the correct mathematical process. The most common mistake was not realising that the mass had to reach 280 mg before the next cup of coffee could be drunk. Some students failed to answer the question being asked and left their answers as 2 hours 56 minutes. This was another example where efficient use of a calculator could have helped some students.

Part (c) was testing the ability of students to criticise the model used in this context. Many students gave one word answers or incomplete sentences. While essays are not required in these types of question, and students should be encouraged to be concise, they do need to make themselves clear as examiners are not permitted to guess what they mean.

Q9.

In part (a) a large proportion of the students recognised the transformation as a stretch although a significant minority incorrectly stated that it was in the x -direction with scale

factor $1/3$. In part (b) there were many correct sketches although it was not uncommon to see the

y – intercept incorrectly given as 1. In part (c), the majority of students scored the first mark by eliminating y correctly to obtain $4^{-x} = 3 \times 4^x$ although a significant minority failed to score even this opening mark because they tried to combine two steps which resulted in the incorrect initial equation, $-x \log 4 = x \log 12$. A significant number of the students failed to make any further progress. More able students who used the log laws correctly to reach $-x \log 4 = \log 3 + x \log 4$ were generally successful in finding the correct value for x . A common error in applying laws of indices, amongst those students who tried to rearrange the equation $4^{-x} = 3 \times 4^x$ before taking logarithms, is illustrated by the incorrect equation

$$16^{2x} = \frac{1}{3}.$$

Q10.

The sketch of the given exponential curve was usually drawn having the correct intercept at $y = 1$, but a significant minority of students did not show its correct behaviour for negative x values – the curve should level out close to the x -axis and certainly nearer to the x -axis than the line $y = 1$. Part (b) was usually answered correctly with almost all showing that they knew how to use logarithms to solve an equation of the form $a^x = b$. The positioning of the minus sign was crucial in part (c) and it was not uncommon to see the incorrect answer $f(x) = -9^x$.

Q11.

This question, which tested the ‘exponentials and logarithms’ section of the specification, resulted in a good spread of marks. Although there were many correct sketches of $y =$

$\frac{1}{2^x}$ seen in solutions to part (a), there were almost as many that were sketches similar to that of $y = 2^x$. Another common incorrect sketch was one which was obtained by reflecting $y = 2^x$ in the x -axis.

Part (b) was generally answered well, although some candidates failed to pick up the final mark because they either left their final answer in the form $-x = 0.322$ or did not give the value of x to at least the required degree of accuracy. Of those who were less successful, some displayed poor algebraic skills or believed incorrectly that cross multiplying led to an equation of the form $10^x = 4$.

Part (c) proved to be demanding, with most candidates failing to obtain the correct expression for y in terms of a and b , although the majority of candidates were awarded partial credit for applying log laws correctly. A typical misconception is illustrated by $2 \log_a b + 3 \log_a y = 5 \log_a by$. Those candidates who were the most successful usually applied no more than one log law (or used $\log_a a = 1$) per line of working to reach, for example, $\log_a b^2 y = 1$ and then stated $b^2 y = a$ before completing by rearranging to obtain $y = ab^{-2}$.

Q12.

Most candidates demonstrated some understanding of the information given in this in-context question on exponential growth and decay. There were many fully correct

solutions.

Most candidates had part (a) correct, with relatively few either not converting 1 hour to 60 minutes or failing to round correctly to the nearest 1000, or not rounding at all.

In part (b)(i), most candidates set up the correct starting-point equation, but they were required to manipulate it to a point where logarithms could be taken before marks were awarded; many failed to do this. Knowledge of using logarithms varied widely. Many were unable to deal properly with the coefficient and the exponential term and wrote down equations in which t actually should have cancelled out. However, many candidates managed to reach a 'solution' from such incorrect working. In contrast, many other candidates gave a concise derivation of the correct answer.

In part (b)(ii), again many candidates set up the correct starting equation, or something close to it, and some used inequalities, which were accepted. Although many did show clearly how the given equation was derived, others could not handle the exponential terms correctly or made sign errors in manipulating their equation. However, many of those who could not derive the quadratic equation in $e^{\frac{1}{3}t}$ did recognise it as such and solved it successfully.

Pleasingly, most candidates also rejected the $e^{\frac{1}{3}t} = -10$ as impossible. Relatively few tried to find a solution again via taking logarithms immediately, but those who did made no progress and abandoned the question.

Q13.

The sketch of the given exponential curve was generally well-done although some students did not display sufficient care in sketching the curve in the negative- x region. In part (b), the majority of students were awarded the mark for correctly eliminating one of the variables, but a significant number could not form a quadratic with any confidence. Those who recognised that 7^{2x} could be written as $(7^x)^2$ usually had no difficulty in proceeding to form and solve a correct quadratic equation in y and then to find the correct value for the y -coordinate of the point of intersection, although a small minority of these students did not give sufficient consideration to the rejection of the inappropriate negative solution. The usual errors seen are illustrated by the following: $7^x = 7^{2x} - 12 \Rightarrow 12 = 7^{2x} - 7^x \Rightarrow 12 = 7^x$ or $7^x = 7^{2x} - 12 \Rightarrow \log 7^x = \log 7^{2x} - \log 12$. There were, however, an encouraging number of completely correct solutions.

Q14.

Most candidates stated, or clearly indicated, the correct values for p , q and r in part (a) and appreciated that these answers could then be used in part (b). There were many correct solutions seen, although a number of candidates resorted to using logarithms rather than just the rules of powers. The most common error in applying laws of indices

was to express $2^{\frac{1}{2}} \times 2^x$ as $2^{\frac{x}{2}}$ rather than $2^{\frac{1}{2}+x}$ which led to the common wrong answer $x = -6$. For full marks in part (b), candidates were required to state the value of x explicitly rather than leave the last line of their working as $2^x = 2^{-3.5}$.

Q15.

Most candidates scored well on this question although some misinterpreted the

information given.

In part (a) most candidates substituted and evaluated correctly. A few, having written the unsimplified form, could not cope with the index and combined the 10 and 2 incorrectly. Some misinterpreted the information and found a mass that decayed to 10 grams in 14 days.

In part (b) most candidates substituted the 16 correctly and then, usually, took $\log$ s in order to

solve for d . Only a minority recognised that $\frac{1}{16} = 2^{-4}$ and so $\frac{d}{8} = 4$. Some candidates began

incorrectly by writing $2^{-\frac{d}{8}} = 16$ which led to a negative answer. Most of these then just dropped the minus sign rather than looking for an error in their working.

In (c) Most candidates set up the correct equation and used logarithms to various bases to solve it. The most common answer here was 53.15, as candidates failed to heed the request to give the answer as an integer. Many also rounded down to 53, misinterpreting what the question was asking.

Not all understood the question and assumed the mass had decayed by 1%, so used 99%.

Q16.

Graphs were drawn correctly by the majority of candidates and the correct vector given for the translation. There are still too many candidates, though, who use alternative words for 'translation' such as "move" or "shift" which are not accepted.

In part (c)(i) the justification of the transformation of the index equation was more of a challenge even to the high grade candidates. A substantial number of candidates simply replaced 4^x with $(2^x)^2$ which gained no credit as it was basically what they had to show. More were successful in showing that $2^{x+2} = 4Y$ by inserting the intermediate stage $= 2^x \times 2^2$. The final part was only done well by better candidates with many misusing the laws of $\log$ s at the outset as illustrated by 'log $4^x - \log 2^{x+2} = \log 5$ '. Of those who did solve the printed quadratic equation in Y , equated the values to 2^x and solved for x , where possible, only a minority gave a conclusion for the "show that" that there was "only one real solution".

Q17.

Candidates' sketches generally displayed a thorough understanding of the graph of $y = 2^x$. In part (b)(i) the trapezium rule was frequently applied correctly and a smaller proportion of candidates than normal failed to give their final answer to the stated degree of accuracy. Part (b)(ii) was almost always answered correctly. In part (c), the question asked for a description of a geometrical transformation. Although answers which gave two separate translations were condoned, the penalty for descriptions which involved two different transformations, for example stretch and translation, was severe. However many correct descriptions were presented, which indicates the continued improvement in candidates' performance on this topic. Part (d), as expected, proved to be demanding. The common wrong approach is illustrated by 'log $y = \log 2^{x+k} + \log 3$ '. Better candidates applied the correct substitution and manipulation to reach $2^k = 5$, but a significant minority

of these candidates, after taking logs and using the log law, left their answer in the form k

$$= \frac{\log 5}{\log 2}.$$

Q18.

Parts (a) and (b) gave the opportunity for all to pick up marks, with the majority giving the two decimal place answer required in part (b). Many candidates scored just one of the two marks in part (c). The translation approach was the more popular, but the fact that the

vector needed to be $\begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$ was too subtle for most, as was the scale factor of $\frac{1}{8}$ for those using a stretch approach. Similarly in part (d), needing $4(x - 1)$ rather than $4x - 1$ defeated the majority. However, this caused no undermining of confidence as many answered part e(i) well, with sufficient justification to support the given answer.

It should be noted, however, that answers which simply started with $k = \frac{8 \times 5}{4}$ with no evidence of using correct log laws could not earn credit. In part (e)(ii) most candidates

knew they had to solve ' $\frac{5}{4} = 2^{4x-3}$ ' and started by taking logs, but then resorted to calculators rather than establish an exact result (with numbers hinted at in the previous part). The best solution, given by some, was to multiply both sides by ' 2^3 ' and then take logs, to the base 10, of ' $10 = 2^{4x}$ ', which led directly to the result.

Q19.

In part (a), many candidates gave the correct value for where the curve met the y -axis, but a significant minority drew no part of the graph in the second quadrant, their graphs stopping at the point (0, 1). Many candidates scored at least half marks for their descriptions of the geometrical transformations in part (b). The most common wrong answer in part (b)(i) was "a stretch of scale factor 2 parallel to the y -axis", and in part (b)(ii) the direction of the translation vector caused the most problems.

Part (c)(i) was poorly answered, with many candidates failing to give sufficient justification for the printed answer. A very common error was to write 9^x as $3y$ and yet still reach the correct factorised quadratic equation.

Although part (c)(ii) was answered better, the majority of candidates failed to gain all the four marks. The most common wrong approach was to take logarithms of each term in the given equation. Those candidates applying this incorrect method had clearly not seen the implication of the "Hence". Those candidates who had seen the link and started with $3^x = 1$ and $3^x = 2$ generally scored all four marks.

Q20.

Candidates generally applied the trapezium rule correctly and gave the final answer to the given degree of accuracy. Many candidates failed to gain full marks in part (b)(ii) as their diagrams did not show four trapeziums. There were also a significant number of candidates who either failed to draw any diagram or who drew the sloping sides of their

trapeziums under the curve. Most candidates realised that the geometrical transformation was a stretch, but the wrong answer 'stretch parallel to the y -axis, scale factor 3' appeared almost as frequently as the correct answer. A high proportion of candidates showed that they were able to use logarithms to solve the equation $6^{3x} = 84$ but some only gave their answer to two decimal places. The final part of the question was poorly answered, although a minority of candidates did score some credit for partially correct answers, for example, $f(x) = 6^{x+1} - 2$.

Q21.

This question also proved to be a good source of marks for a majority of candidates. Normally part (a) was answered correctly but there were also signs of poor examination technique. Candidates who just wrote down the y -coordinate of P as '3' scored no marks but those who wrote ' $3(2^0 + 1) = 3$ ' scored 1 of the 2 marks available. Responses to the trapezium rule question were mixed with many candidates scoring full marks but a greater number of candidates failing to pick up even the method mark. The two main method errors involved either the use of seven ordinates rather than the required four or the use of $x = 0, 1, 2, 3$ only.

In part (c)(i) a significant number of candidates were unable to gain the mark because they wrote 3×2^x as 6^x . Many candidates gained the 2 method marks for using logarithmic rules but a significant minority failed to round 2.5849 correctly to three significant figures. The wrong answer 2.59 was seen more than expected.

Q22.

The evaluations in part (a) were done correctly by most candidates, although some made numerical errors. In part (a)(ii), the request for three significant figures was largely adhered to, although some interpreted this as three decimal places.

Candidates started part (b) well but many were confused as to how to handle the negative signs.

Many attempts to take logs of negative numbers were seen. Many got to $14 \ln\left(\frac{5}{12}\right)$ or $-14 \ln\left(\frac{5}{12}\right)$ and gave this as their answer. Despite this, the vast majority gave the correct answer of 12 to part (b)(ii), some just ignoring their negative sign or saying it must be positive. They apparently did not respond to a negative answer by reviewing their answer to part (b)(i). Very few candidates explained what they were doing with the negative index to obtain the expected answer. In part (c)(i), very few candidates derived the differential equation with confidence. Some made a start by attempting to differentiate the expression $\frac{1}{14}$ for x , although, of those, many made the error of bring down the t together with $\frac{1}{14}$. Others attempted to find an expression for t and differentiate that. There was usually little progress beyond these starting points, although several candidates attempted to convince the examiner that they had found the requested result, with much abuse of the $\ln$ function seen in such attempts.

Some tried to integrate the given differential equation to recover the expression in x , and some very neat solutions were seen this way, although those who did not consider limits

or a constant could make no further progress. Again, handling negative signs with logarithmic expressions caused difficulties. Most candidates scored the final mark for

applying the result from part (c)(i), although some did say that $\frac{1}{14}(7) = 2$.

Q23.

A majority of candidates indicated that the method for solving equations of the form $a^x = b$ was well known. The most common errors are illustrated by

(a) $\log x^{0.8} = \log 0.05$

(b) (i) $x = \frac{\log 0.8}{\log 0.05}$

- (ii) answers 13.4 or 13.43 Part (b), which tested geometric series, although answered much better than the corresponding question in summer 2005, still indicates a general weakness in this area of the specification and the associated algebraic manipulation. It was not uncommon to see candidates using a specific value of a , either 1 or 20 or even 100, to solve the equation

$\frac{a}{1-r} = 5a$ In the final part a significant proportion of the candidature used the formula for the sum to n terms rather than the formula for the n th term. Some better candidates obtained $n = 14.5\dots$, but failed to realise that n was an integer.

Q24.

- (a) The correct value of a was often seen, but $a = -9$ was a common incorrect answer. The majority of candidates gained the method mark in finding b , but a number of candidates then continued with incorrect manipulation of the exponential function.
- (b) The majority of candidates answered this correctly.

Q25.

- (a) Reasonable attempts were made by most candidates. The main mistake was finding the derivative of x^{-2} which was often seen as $-2x^{-1}$. This lost marks since candidates then 'found' the answer given by incorrect means.
- (b) There were many fully correct answers but many candidates chose incorrect functions for u and $\frac{dv}{dx}$, with the obvious implications.
- (c) (i) This was reasonably well answered, the main error seen being $1 - 21nx = x^3$.
- (ii) The majority of candidates obtained the method mark following through from their answer to part (b). Few candidates completed this question correctly.

Q26.

This question was answered very well with most candidates gaining nearly full marks. In part (a) many wrote £80 instead of just 80 but this was accepted.

Part (b) was mostly done accurately with many getting to $k = \sqrt[56]{62.5}$ which was accepted as *showing that*. Those who used logarithms lost a mark if they didn't show how $\log 62.5 = 56 \log k$ led to the given value of k .

Virtually all candidates got part (c) correct, with both the approximate answer 200707 and the more exact 200648 being seen, often with several decimal places attached, but this was of no matter as candidates had demonstrated a value greater than £200,000. In part (c)(ii) there were occasional errors although most candidates used logarithms to derive 124.7 correctly whilst a few used trial and improvement to deduce 125. However, many lost the last mark in incorrect interpretation of the year, or through not actually giving a year. The year 2025 was the common wrong answer.

Q27.

This question proved to be a good source of marks for many candidates. Most candidates answered both parts of (a) correctly although the usual errors, $3^\circ = 0$ or $3^\circ = 3$ and $3^X = 27$

$\Rightarrow x = \sqrt[3]{27}$ were seen. The trapezium rule, required in part (b), was generally well understood although some candidates are still mixing up 'ordinates' and 'strips' or failing to use sufficient brackets correctly. Although many candidates scored well in parts (c) there were others in (i) who used the relevant law of logarithms incorrectly or others who failed to indicate that logarithms had been used at all. In (ii) not all candidates realised that a simple substitution of 13 for 3^X in $k = 27 - 3^X$ with evaluation was all that was required. Descriptions of transformations continue to pose a problem for candidates. Many candidates gained partial credit for their sketches but both marks were rarely awarded.

Q28.

Part (a) This was very poorly answered by the majority of candidates. Range is a topic that is not very well understood by candidates.

Part (b) This was reasonably well answered. Errors occurred where candidates, on interchanging x and y , tried to work with $x = 2e^{3y} - 1$.

Part (c) This was poorly answered by most of the candidates. Many candidates obtained the correct differentiation of the $\ln$ function but lost marks because they omitted to multiply by $1/2$. This resulted in the very common incorrect answer of $2/3$.

Q29.

Many candidates incorrectly evaluated $(3^0 + 1)$ as 1 in part (a) although surprisingly they frequently recovered and used the correct answer, 2, in part (b). In general, the trapezium rule was well understood, although it was relatively common to find an arithmetical slip in evaluating $(3^x + 1)$; the '1' was not added in one of the terms. Candidates should have given the final answer to the degree of accuracy specified in this numerical integration question. Some of the weaker candidates just integrated $(3^x + 1)$ directly as $(3^{x+1} + x)$. The use of a diagram to show that the approximation is an overestimate was well understood, although some used rectangles rather than the required trapeziums and others had the

sloping sides of their trapeziums in the wrong positions, that is, below the curve. Part (c) was generally well answered although some weaker candidates tried to take logarithms of each term rather than to rearrange the equation into the form ' $3^x = 4$ ' as a first step. Those candidates who used trial and improvement usually failed to produce an acceptable solution. In part (d), a common wrong expression for $f(x)$ was " $-3^x + 1$ ", although a significant minority did obtain the correct answer " $3^{-x} + 1$ ".

Q30.

Part (a) tested the relationship of e^x and $\ln x$. This part was well answered. Part (b) required algebraic manipulation using e^{-x} which discriminated between candidates. The final part was accessible to candidates who failed on the previous part. Candidates were required to give exact answers and not decimal equivalents.



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